

Introduction



- Black holes pull in gas/dust/other debris from its surrounding area
- Some “blobs” of this gas can emit light as the accretion disc spins in the form of X-ray flickers
- Blobs can tell us about physical characteristics of the black hole

WHY?

1. Lack of a streamlined end-to-end model that can validate the existence of QPOs without specific case studies
2. Most literature focuses on specific types of QPOs but doesn't seek to generalize across classifications
3. Takes time + qualitative analysis to validate QPOs

HOW?

Literature has shown that the incorporation of the following 3 factors is vital in QPO analysis.

1. Inclusion of Poisson Noise: This accounts for randomness in the way that photons hit detectors from satellites
2. Signal Superimposition: Mimics the “quasi-periodic” nature of these signals
3. Cross-phase power spectrum lag: Accounts for discrepancies across photon energy levels from the same region of the body

Purpose

Research Question: Can an end-to-end synthetic waveform model be used to model QPOs as a whole?

Prediction & Hypothesis: An end-to-end synthetic waveform model will be able to accurately validate QPO existence due to the inclusion of Poisson noise integration, signal superimposition, and cross-phase power spectrum lag that provide evidence to support the existence of QPOs in specific regions

Independent Variable: Testing the accuracy of a specific model, so that would be the independent variable here.

Dependent Variable: Accuracy of the model in validating QPO existence

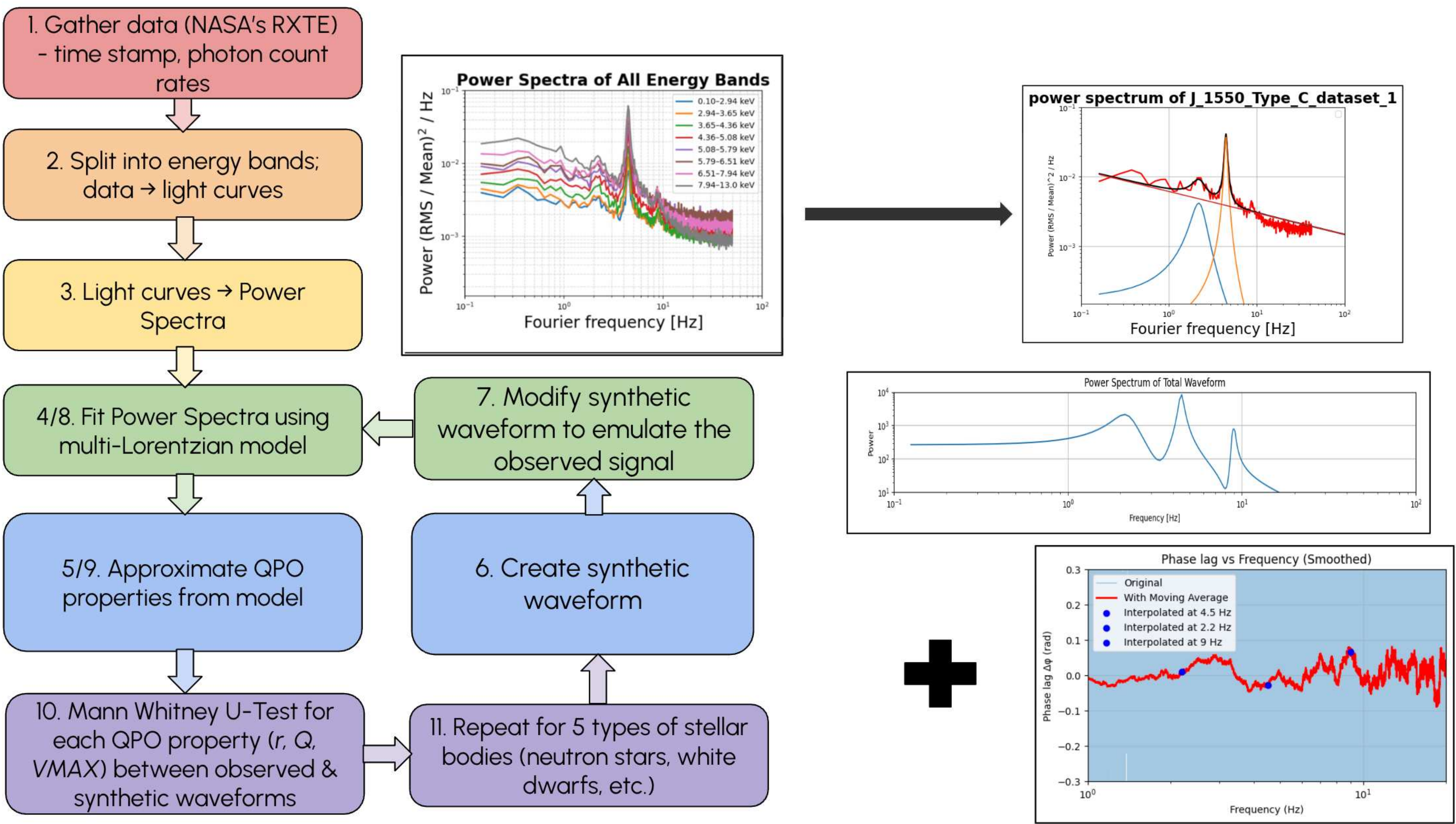
Constants: Observed waveform analysis technique and overall methodology skeleton

Controls: N/A

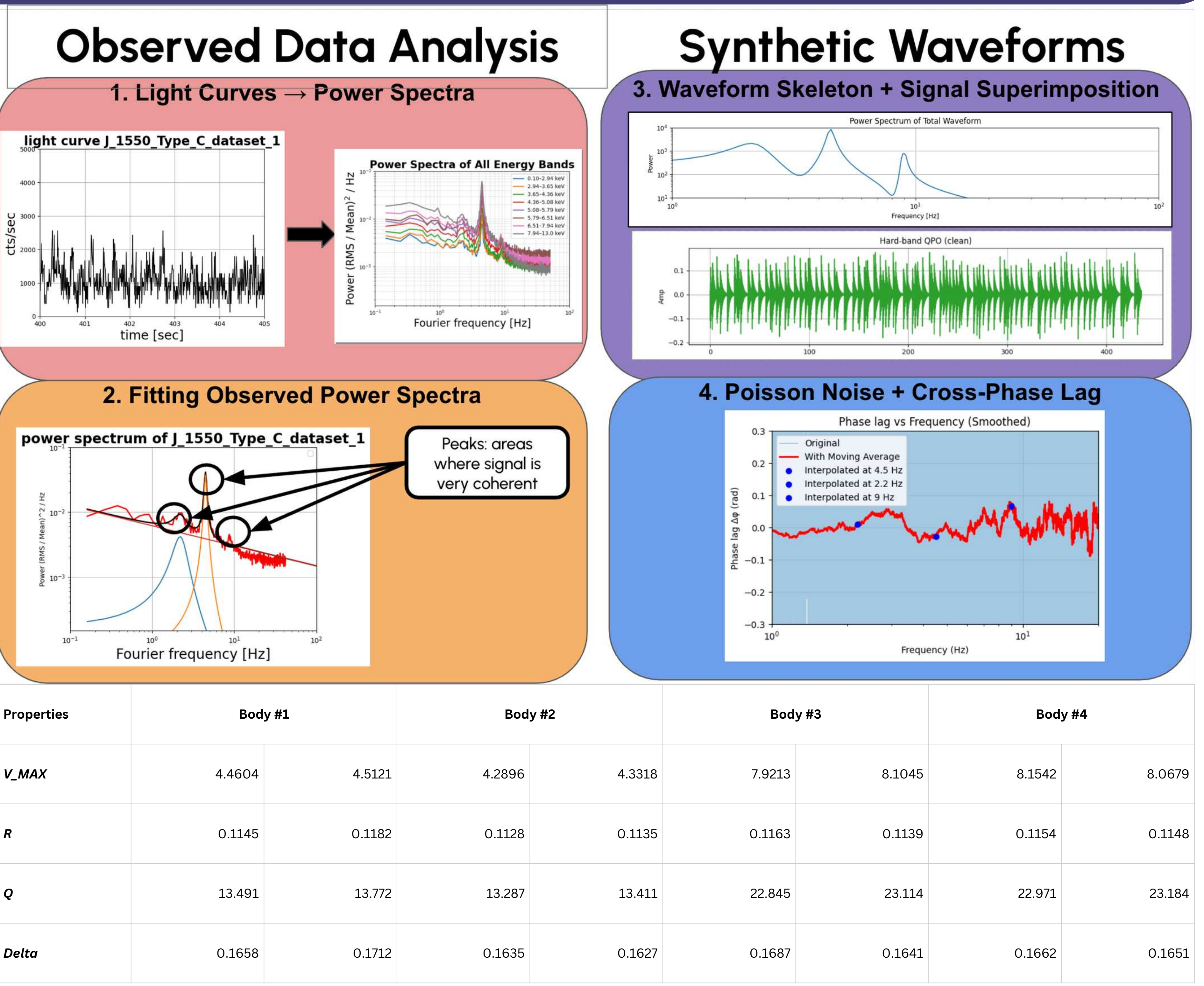
Validating the Existence of Quasi-Periodic Oscillations in Astronomical Objects Through Synthetic Waveform Creation

All images were created by the author unless otherwise specified

Methodology



Data/Results



Conclusions

Overall, the synthetic waveform model performed well in validating QPO properties. The model appeared to work fairly well across all QPO properties, with the Mann-Whitney U tests for each body coming up with a p-value < 0.05, signifying that differences are likely due to random chance. a

Bodies	P-Value (α = 0.05)
Body #1	0.0333
Body #2	0.0002
Body #3	0.001
Body #4	0.0427

Limitations/Future Research

Even though the synthetic waveform model performed well for the tested bodies, this is only a small number of total bodies that have been observed. Moreover, only black holes were tested as opposed to an array of astronomical objects. This is something that could be researched in future projects. The model also likely has properties that would not necessarily work for neutron stars or other bodies.

References

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